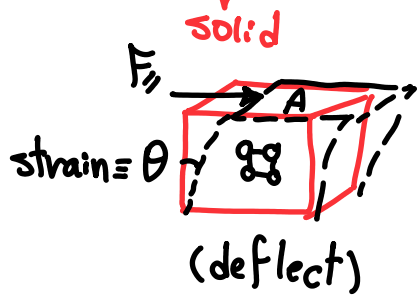
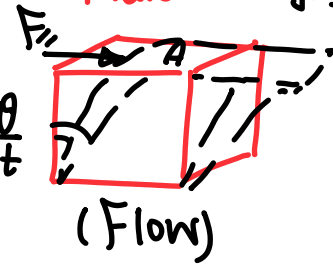


Fluid Properties

Matter

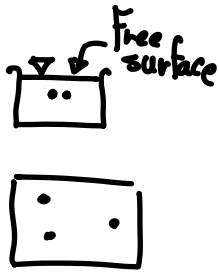


rate of shear strain $\equiv \frac{d\theta}{dt}$



Liquid

gas



Fluid study

Fluid at rest
(Fluid statics)

Fluid at motion

(Fluid kinematics)

(Fluid dynamics)

Units & Dimensions

Units

SI units

mass $\equiv \text{Kg}$

length $\equiv \text{m}$

time $\equiv \text{sec}$

Cgs units

mass $\equiv \text{gm}$

length $\equiv \text{cm}$

time $\equiv \text{sec}$

Dimensions

MLT system

M $\equiv$ mass

L $\equiv$ length

T $\equiv$ Time

Fluid properties:-

1- Density $\equiv \rho$:- mass per unit volume

$$\rho = \frac{m}{V} \quad [\text{Kg/m}^3], [ML^{-3}]$$

$$\rho_{\text{std. liquid}} = \rho_{\text{H}_2\text{O @ 4}^\circ\text{C}} = 1000 \text{ Kg/m}^3$$

$$\rho_{\text{mercury}} = \rho_{\text{Hg}} = 13600 \text{ Kg/m}^3$$

$$\rho_{\text{std. gas}} = \rho_{\text{Air @ 1 bar, 298K}} = 1.18 \text{ Kg/m}^3$$

2- Specific Volume $\equiv v$:- Reciprocal of density

$$v = \frac{1}{\rho} = \frac{V}{m} \quad [\text{m}^3/\text{Kg}], [M^{-1} L^3]$$

$$N = \frac{\text{Kg m}}{\text{s}^2}$$

3- Specific Weight $\equiv \gamma$:- weight per unit volume

$$\gamma = \frac{W}{V} = \frac{mg}{V} = \rho g \quad [N/\text{m}^3] \equiv \left[\frac{\text{Kg m}}{\text{s}^2} \frac{1}{\text{m}^3} \right]$$
$$\equiv [\text{Kg/m}^2 \text{s}^2], [M L^{-2} T^{-2}]$$

Esham siefs

4- Specific gravity $\equiv$ S.G. $\cdot$ Relative density $\equiv \rho$ (Unitless) $\cdot$ (Dimensionless)
 $M^0 L^0 T^0 \equiv 1$

For Liquids
 $S.G._{liq} = \frac{\rho_{liq}}{\rho_{std. liq}} = \frac{\rho_{liq}}{\rho_{water}} = \frac{\rho_{liq}}{1000}$

For Gases
 $S.G._{gas} = \frac{\rho_{gas}}{\rho_{std gas}} = \frac{\rho_{gas}}{\rho_{Air @ 1bar, 298K}} = \frac{\rho_{gas}}{1.18}$

5- Viscosity $\equiv$ dynamic viscosity $\equiv$ absolute viscosity $\equiv$ Coef of Viscosity $\equiv \mu$:-
 Fluid resistance For shear stress (Flow)

Newton's Law of Viscosity:-

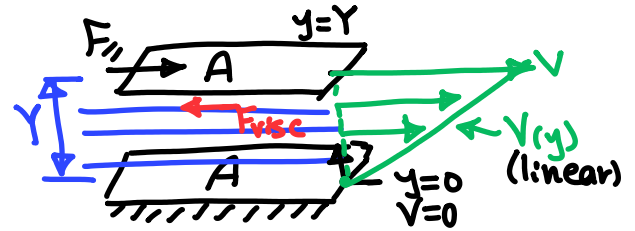
$A \equiv$ Contact Area (m^2)

$Y \equiv$ gap between Plates (m)

$F_{||} \equiv$ shear Force $\equiv$ Drag Force (N)

$\tau \equiv$ shear stress $= \frac{F_{||}}{A}$ (N/m^2) $\equiv$ (Pa)

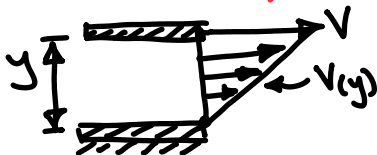
$V \equiv U \equiv$ Constant Velocity (m/s)



$$\vec{F}_{||} = \vec{F}_{vis} = A \mu \frac{\Delta V}{\Delta y} = A \mu \frac{V-0}{Y} = A \mu \frac{V}{Y}$$

Velocity Profile

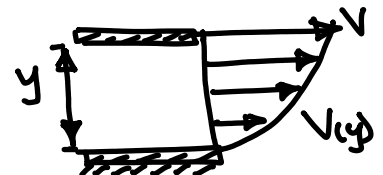
Linear (Default)



$$F_{vis} = F_{||} = A \mu \frac{\Delta V}{\Delta y} = \text{Const}$$

$$\tau = \mu \frac{\Delta V}{\Delta y} = \text{Const}$$

Non Linear



$$F_{vis} = F_{|| (y)} = A \mu \frac{dV}{dy} \neq \text{Const}$$

$$\tau_{(y)} = \mu \frac{dV}{dy} \neq \text{Const}$$

General Newton's Law of Viscosity:-

$$\vec{F}_{||} = \vec{F}_{vis} = A \mu \frac{dV}{dy}$$

Units of viscosity (dynamic viscosity) $\equiv \mu$

SI unit of μ :-

$$\begin{aligned} \mu &= \frac{\tau \Delta y}{\Delta V} = \frac{Pa \cdot m}{\frac{m}{s}} = Pa \cdot s \equiv \frac{N}{m^2 \cdot s} \\ &= \frac{kg \cdot m}{s^2} \cdot \frac{1}{m^2} \cdot s = [kg/m \cdot s] \\ &= [M L^{-1} T^{-1}] \end{aligned}$$

C.g.s unit of μ :

$$1 \text{ gm/cm s} = 1 \text{ Poise} = \frac{1}{10} \text{ Pa.s}$$

$$1 \text{ Centi Poise} = 10^{-2} \text{ Poise} = 10^{-3} \text{ Pa.s}$$

G- Kinematic viscosity $\equiv \nu$:

$$\nu = \frac{\mu}{\rho}$$

SI unit of ν :

$$\nu = \frac{\frac{\text{kg}}{\text{m s}}}{\frac{\text{kg}}{\text{m}^3}} = \text{m}^2/\text{s} \quad [L^2 T^{-1}]$$

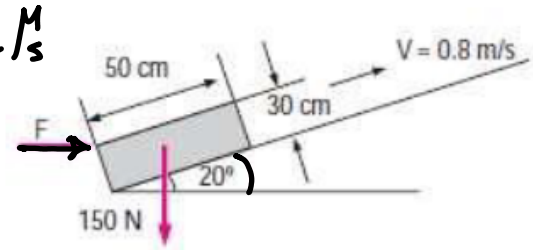
c.g.s unit of ν :

$$\nu = 1 \text{ cm}^2/\text{s} = 10^{-4} \text{ m}^2/\text{s}$$

$$1 \text{ Stokes} \equiv 1 \text{ cm}^2/\text{s} = 10^{-4} \text{ m}^2/\text{s}$$

sheet (1)

12- A 50-cm x 30-cm x 20-cm block weighing 150 N is to be moved at a constant velocity of 0.8 m/s on an inclined surface with a friction coefficient of 0.27. (a) Determine the force F that needs to be applied in the horizontal direction. (b) If a 0.4-mm-thick oil film with a dynamic viscosity of 0.012 Pa.s is applied between the block and inclined surface, determine the percent reduction in the required force.

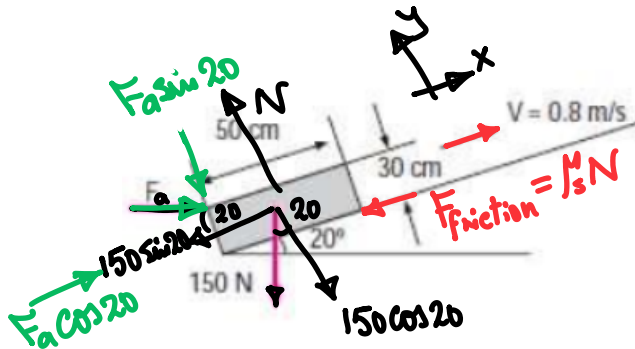


Case (A): solid Friction F_a

$$\sum F_x = 0$$

$$F_a \cos 20 - 150 \sin 20 - \frac{1}{5} N = 0$$

$$F_a \cos 20 - 150 \sin 20 - 0.27 N = 0 \quad \text{--- (1) } N \propto F_a$$



$$\sum F_y = 0$$

$$N - 150 \cos 20 - F_a \sin 20 = 0 \quad \text{--- (2) } N \propto F_a$$

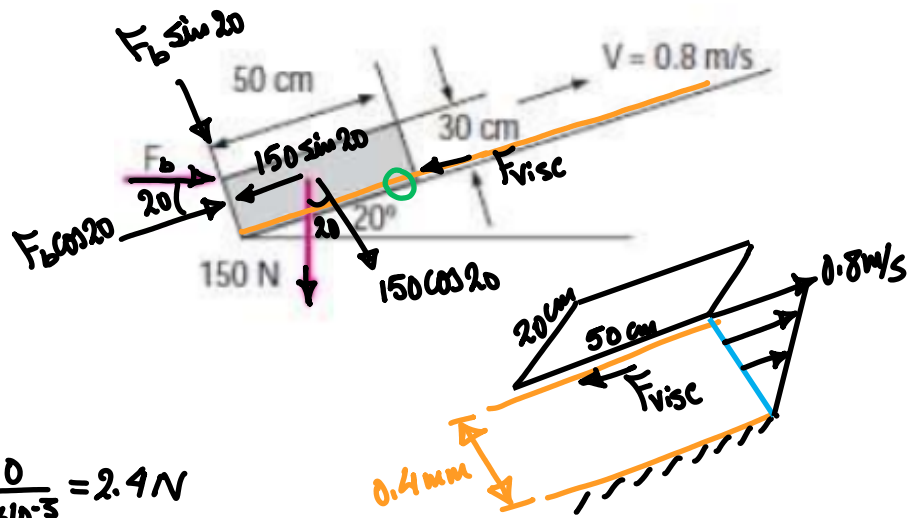
$$\rightarrow N = 150 \cos 20 + F_a \sin 20$$

substitute in (1):

$$\therefore F_a \cos 20 - 150 \sin 20 - 0.27(150 \cos 20 + F_a \sin 20) = 0$$

$$\therefore F_a = 105.45 \text{ N}$$

Case (B):- Viscous Friction:- F_b



$$F_{visc} = \mu A \frac{\Delta V}{\Delta y}$$

$$= 0.012 * 50 * 20 * 10^{-4} * \frac{0.8 - 0}{0.4 * 10^{-3}} = 2.4 \text{ N}$$

$$\sum F_x = 0$$

$$F_b \cos 20 - 150 \sin 20 - F_{visc} = 0$$

$$F_b \cos 20 - 150 \sin 20 - 2.4 = 0 \quad \therefore F_b = 57.15 \text{ N}$$

$$\% \text{ reduction} = \frac{F_a - F_b}{F_a} * 100 = \frac{105.45 - 57.15}{105.45} * 100 = 45.57\%$$